

## RESEARCH ARTICLES

## SOLAR CELLS

# Plasma surface engineering for efficient and stable perovskite solar cells and modules

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The instability of perovskite solar cells (PSCs) stems largely from the formation and evolution of interfacial defects associated with the soft, multicomponent perovskite lattice. We report a scalable, plasma-based passivation strategy that forms conformal, uniform, and strong-bonded heterostructure through in situ chemical reactions on large-area perovskite films. This approach also mitigates defect accumulation within the laser-scribed interconnection regions, where localized damage often dominates module-level performance losses. We achieved a power conversion efficiency (PCE) of 27.2% in small-area devices (active area 8.313 square millimeters) and 24.0% (certified efficiency of 23.5%) in 100-square-centimeter (cm<sup>2</sup>) modules (aperture area 65.05 cm<sup>2</sup>). The small-area device retained 98.1% of its initial PCE after 2000 hours of maximum power point tracking at 85°C under 1-sun illumination, and the 100-cm<sup>2</sup> module retained 99.3% of its initial PCE after 1600 hours at 65°C under 1-sun illumination.

Perovskite solar cells (PSCs) can offer low cost and mechanical flexibility (1–4), but challenges in achieving long-term operational stability still limit their potential commercialization. Unlike conventional crystalline silicon, perovskites (PVSKs) have intrinsically low formation energies and soft lattices composed of large ions with low valence. Moreover, PVSK films are typically processed in solution, where rapid supersaturation-driven nucleation and heterogeneous crystallization lead to polycrystalline microstructures with abundant grain boundaries and compositional inhomogeneity (5, 6).

These structural characteristics, combined with the intrinsically soft lattice of PVSK, lower the energy barriers for defect formation and ion migration and facilitate formation of complex, multiscale defects that ultimately degrade carrier lifetimes and compromise long-term device stability. Notably, the surface defect densities of the PVSK film typically exceed those in the bulk by one to three orders of magnitude, making the interfacial region a critical location for nonradiative recombination, ion migration, and accelerated absorber degradation (7, 8). Thus,

eliminating interfacial defect-rich layers and constructing intrinsically stable heterointerfaces with efficient charge transport are essential (9, 10).

Interfacial defect states in PVSK devices can be mitigated through various surface-remediation and passivation strategies. Solid-phase techniques, such as adhesive-tape peeling and laser polishing, have been developed to physically eliminate defect-rich surface layers (11, 12). These solvent-free methods avoid chemical damage and enable mechanical transfer, but it remains difficult to achieve multiscale compact passivation across rough surfaces because of limited solid-solid contact (8). More commonly, solution-phase chemical passivation (13–16) with inorganic salts, organic small molecules, and functional polymers has been used to improve surface electronic properties, suppress deep-level traps, and enhance optoelectronic performance by filling vacancies, neutralizing charges, or forming Lewis acid-base complexes (17, 18). However, the uncontrolled volatility of solvents triggers evaporation-driven concentration fluctuations of passivation molecules, so this method can create heterointerfaces with poor spatial uniformity, which becomes more severe when scaling up for large-area modules (19, 20). Additionally, new interfaces in modules, such as laser-scribed interconnection zones (also referred to as heat-affected zones or HAZs), are difficult to passivate with conventional, solution-based methods (21–23).

Recently, vapor-phase passivation using highly reactive gases has been explored for PVSK modules (PSMs), offering high uniformity and acceptable scalability. Techniques based on heating solid and liquid precursors to generate vapor such as HF (19), 2-phenylethylamine, ethylenediamine (24), and fluorobenzenethiol (25) can passivate large areas. However, practical implementation remains challenging owing to the difficulty of precisely controlling gas-phase concentrations, which are intrinsically coupled to precursor sublimation or volatilization rates. The high reactivity of these vapors also raises substantial environmental and toxicity concerns. Thus, a passivation strategy capable of precise heterostructure construction while simultaneously achieving multiscale compactness, high spatial uniformity, and practical scalability—particularly across the intricate interfaces of PSMs—remains a critical bottleneck.

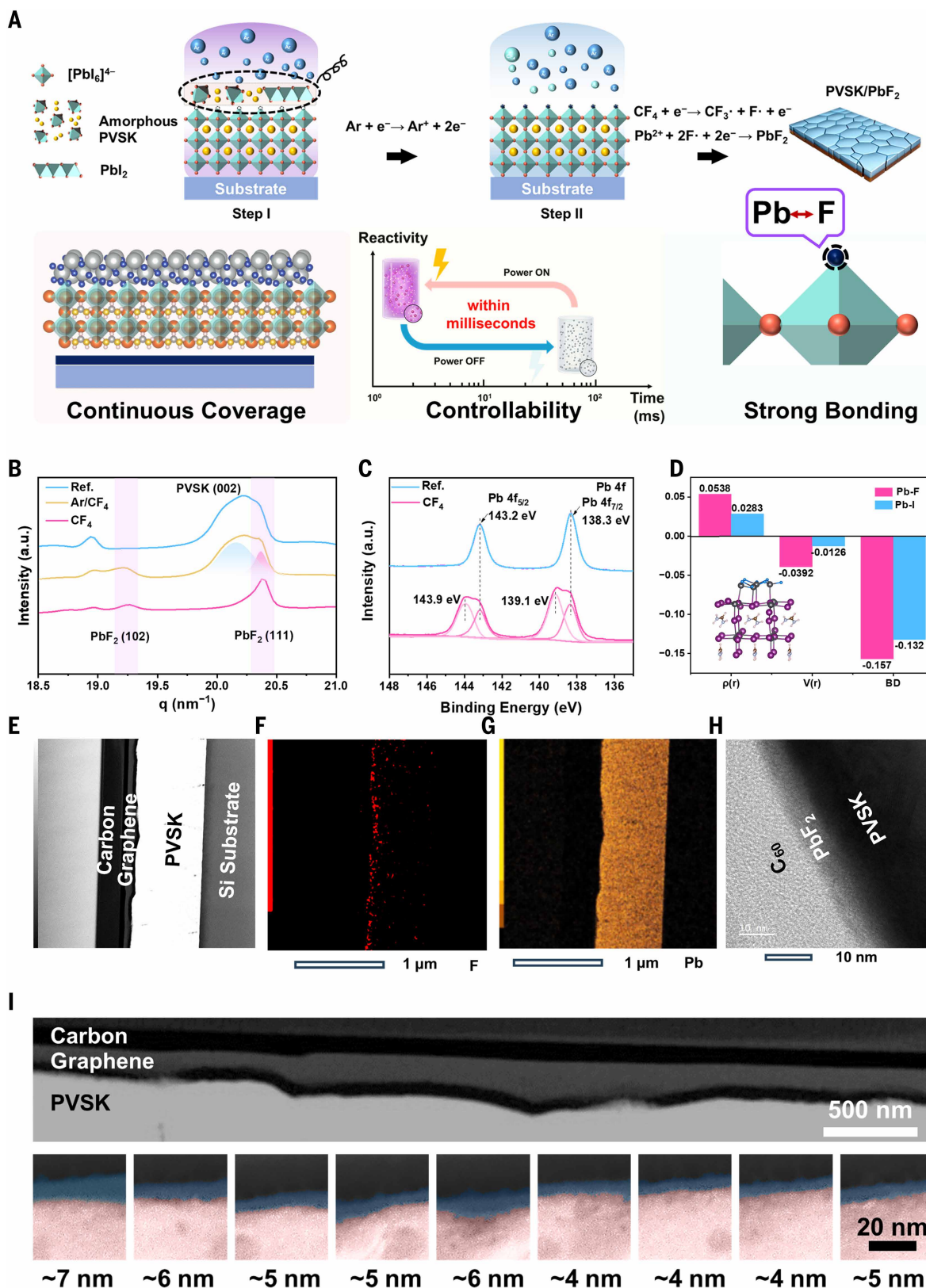
We, on the basis of pioneering works (table S1), report a plasma-based, industry-compatible strategy utilizing electrically gated reactive species to achieve atomic-scale etching and in situ interfacial reaction that creates a uniform, strong-bonded passivation layer on common PVSK components such as FA<sub>0.95</sub>CS<sub>0.05</sub>PbI<sub>3</sub> across large-area films. The obtained fluorinated interface could effectively passivate surface defects (from  $6.63 \times 10^{16}$  to  $5.01 \times 10^{15}$  cm<sup>-3</sup>) through Pb–F coordination while preserving the structural and electronic integrity of the bulk lattice. Multiscale spatial mapping confirmed exceptional uniformity from across the nanoscale to the macroscale. When we applied this process to the HAZs to remediate defect states, it led to an approximately twofold increase in average photoluminescence (PL) intensity and reduced module-specific interfacial energy loss.

The resulting devices achieved a power conversion efficiency (PCE) of 27.2% for small-area cells (0.08313 cm<sup>2</sup>) and 24.0% for 100-cm<sup>2</sup> PSMs (aperture area 65.05 cm<sup>2</sup>). The devices with stable structure could retain 98.1% of their initial efficiency after 2000 hours of continuous operation under 1-sun illumination at 85°C for small-area solar cells and retain 99.3% after 1600 hours at 65°C for 100-cm<sup>2</sup> modules (aperture area 65.05 cm<sup>2</sup>). The devices also showed no obvious performance loss under 12-hour light/12-hour dark diurnal cycling (3000 hours), reverse bias (–14.3 V), and ultraviolet (UV) irradiation (117.6 kWh m<sup>-2</sup>), respectively.

## Plasma surface treatment and PbF<sub>2</sub> film structure

The plasma-based in situ surface treatment technique used a mix of Ar and CF<sub>4</sub> (Fig. 1A and fig. S1). During the treatment, Ar ions primarily served as physical etching agents that polished the PVSK defect-rich

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**Fig. 1. Conceptual design and preliminary validation of the plasma surface treatment process.** (A) Schematic illustration of the plasma-based surface treatment technique. (B) GIWAXS intensity profiles for the PVSK films without plasma treatment, including Ar/CF<sub>4</sub> mixture plasma treatment as well as pure CF<sub>4</sub> plasma treatment. (C) XPS spectra of Pb 4f signal for PVSK films with and without plasma treatment. (D) Electron density  $\rho(r)$ , the potential energy density  $V(r)$  and bond degree (BD) for the Pb–F and Pb–I bond, respectively. (Inset) Structures of FAPbI<sub>3</sub> and PbF<sub>2</sub>. (E) Cross-sectional TEM image and corresponding (F) F and (G) Pb elemental mapping of the plasma-treated PVSK film. The sample was prepared with a structure of Si substrate, PVSK, C<sub>60</sub>, graphene capping layer, and carbon. (H) TEM image of the PVSK/PbF<sub>2</sub>/C<sub>60</sub> interface. (I) Cross-sectional TEM images of the plasma-treated PVSK surface. The enlarged images were randomly selected from the PVSK top surface.

surface (corresponding to Step I of the plasma surface treatment) (supplementary text S1), whereas  $\text{CF}_4$  generated fluorine-containing radicals such as  $\text{F}\cdot$  and  $\text{CF}_3\cdot$  that reacted with uncoordinated Pb atoms to form a strongly bonded passivation layer (corresponding to Step II). In the following discussion, the untreated samples are denoted “Reference” or “Ref.,” whereas the plasma-treated samples are denoted “Target.”

Highly reactive fluorine-containing radicals were generated exclusively under high-voltage plasma activation (5 W radio frequency). After the applied voltage was removed, these radicals rapidly annihilated within several milliseconds and terminated in situ surface passivation reaction (26). The detailed experimental procedure was described in the supporting information, where the PVSK film was subsequently exposed to an Ar plasma atmosphere of 40 standard cubic centimeters per minute (scm) for 2 s and an Ar/ $\text{CF}_4$  mixture of 140/2 scm for 3 s as the optimized condition for film fabrication. The plasma equipment was placed under ambient conditions without additional humidity control, and the chamber was evacuated to around  $10^{-4}$  Pa before plasma processing to minimize residual water influence.

We used complementary microscopy, diffraction, and spectroscopy techniques to assess how plasma processing affected the structure and morphology of PVSK films. Relative to the pristine PVSK film, Ar plasma treatment introduced no discernible additional x-ray diffraction (XRD) peaks (fig. S14). However, after Ar/ $\text{CF}_4$  plasma treatment for 30 s, an additional diffraction peak appeared at  $29.1^\circ$  that we attributed to the (111) reflection of  $\text{PbF}_2$  (fig. S15). Grazing incidence wide-angle x-ray scattering (GIWAXS) measurements also revealed  $\text{PbF}_2$  (102) and (111) diffraction peaks on the PVSK surface region after Ar/ $\text{CF}_4$  plasma treatment (Fig. 1B and fig. S16). Upon replacing the Ar/ $\text{CF}_4$  mixture plasma with pure  $\text{CF}_4$  plasma to treat the pristine PVSK film, the (002) peak intensity of PVSK at scattering vector  $q = 20.1 \text{ nm}^{-1}$  was reduced by an order of magnitude, whereas the intensity of the representative signal peak at  $q = 20.4 \text{ nm}^{-1}$ , corresponding to the (111) peak of  $\text{PbF}_2$ , increased by  $\sim 30\%$ . These changes confirmed the plasma-induced, in situ conversion of PVSK into  $\text{PbF}_2$ . On the basis of a Scherrer analysis, we estimated that the  $\text{PbF}_2$  crystallite size was  $\sim 50 \text{ nm}$  after extended Ar/ $\text{CF}_4$  plasma exposure for 30 s. Assuming that  $\text{PbF}_2$  growth scaled linearly with the exposure time, the corresponding growth rate in Step II could be estimated as  $\sim 1$  to  $2 \text{ nm s}^{-1}$ .

To further demonstrate the formation of  $\text{PbF}_2$ , we deposited the PVSK on a copper grid and treated it with plasma. The transmission electron microscope (TEM) images of the PVSK grains revealed two types of lattice fringes with interplanar spacing of 0.64 and 0.29 nm, respectively, which we indexed to the (100) plane of  $\text{FAPbI}_3$  (27) and the (111) plane of  $\text{PbF}_2$  (fig. S17, A to C), respectively. Localized energy-dispersive x-ray spectroscopy (EDX) characterization also confirmed that the F signal was uniformly distributed across the sample surface on the nanoscale (fig. S17, D to F). To macroscopically verify fluoride incorporation, a lateral heterostructure was engineered on the PVSK film, using masking tape to create a distinct boundary between  $\text{CF}_4$ -treated and pristine regions. Line-scan EDX showed a more than two-fold increase in the F signal in the plasma-treated region (fig. S18).

To elucidate the spatial distribution of the heterostructure, PVSK films were deposited on silicon substrates and then underwent  $\text{CF}_4$  plasma treatment, where cross-sectional lamellae were produced using focused ion beam (FIB) milling. EDX mapping of the cross sections revealed accumulation of fluorine (F) at the top surface of the PVSK layer (Fig. 1, E to H). High-magnification TEM imaging of FIB-prepared lamellae revealed a continuous interfacial layer with a thickness of 4 to 7 nm on the PVSK surface (Fig. 1I). These observations confirmed the effective coverage of the  $\text{PbF}_2$  layer on the PVSK film.

Time-of-flight secondary ion mass spectrometry measurements, including both lateral and depth-resolved analyses, showed pronounced enrichment of F at the top surface in the Target films ( $\sim 10^4$  counts) compared with the Ref. sample ( $\sim 10^2$  counts, background noise) (fig.

S19). The O signal, used to identify the indium tin oxide (ITO) interface, rose only after  $\sim 350 \text{ s}$  of sputtering, whereas the F signal decayed to the background signal within  $\sim 10 \text{ s}$ , indicating that fluorine was predominantly localized at the PVSK surface. These characterizations demonstrated the successful formation of the PVSK/ $\text{PbF}_2$  heterostructure on the top surface.

To gain insights into the chemical states and bonding environments of the heterostructure, PVSK films were treated with pure  $\text{CF}_4$  plasma and characterized by x-ray photoelectron spectroscopy (XPS). A distinct F 1-s signal at 683.8 eV on the Target confirmed that  $\text{CF}_4$  plasma effectively fluorinated the PVSK surface. Additionally, the pristine PVSK film showed peaks at 143.2 eV (Pb 4f<sub>5/2</sub>) and 138.3 eV (Pb 4f<sub>7/2</sub>). After surface fluorination, additional features emerged at 143.9 and 139.1 eV, indicating a modified Pb chemical environment arising from F–Pb bonding, which was consistent with the reported characteristic peak positions for Pb 4f in  $\text{PbF}_2$  (28, 29) (Fig. 1C).

Compared with the Ref. PVSK film (I 3d<sub>3/2</sub> at 630.5 eV and I 3d<sub>5/2</sub> at 619.1 eV), the Target film exhibited a slight shift of the I 3d signal to higher binding energies (I 3d<sub>3/2</sub> at 630.7 eV and I 3d<sub>5/2</sub> at 619.2 eV). This shift could have resulted from the surface fluorination, in that the strongly electron-withdrawing F species can draw electron density away from the Pb–I framework toward the F atoms and make iodine electron-deficient (fig. S20).

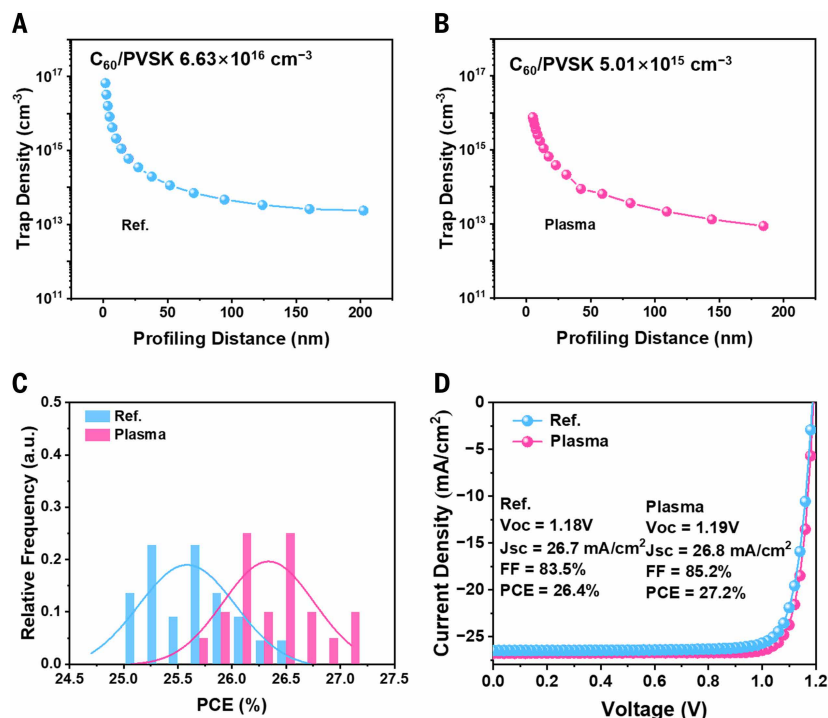
We used density functional theory (DFT) to examine the nature of the Pb–X bonds in surface-fluorinated PVSK film (Fig. 1D and fig. S21) and to quantify the bonding features at the bond critical points (CPs) using the atoms-in-molecules (AIM) framework (table S2) (30), where  $\text{PbF}_2$  clusters were used on the PVSK surface and fully relaxed to mimic the interfacial nonperiodic contact state. The Pb–I/Pb–F interaction was intrinsically polar, exhibiting mixed covalent-ionic character (31, 32). The negative total energy density  $H(r)$  at the CP (33) together with the positive  $\eta$  index (34) indicated a substantial covalent contribution. Notably, the Pb–F bond had a more negative bond degree (BD) ( $-0.157$  for Pb–F and  $-0.132$  for Pb–I), suggesting a higher bond strength when compared with the Pb–I bond. Increased magnitudes of the electron density  $\rho(r)$  (0.0538 for Pb–F and 0.0283 for Pb–I) and the potential energy density  $V(r)$  ( $-0.0392$  for Pb–F and  $-0.0126$  for Pb–I) at the CP for Pb–F were also seen.

## Defect states and electronic properties

To evaluate the influence of plasma treatment on surface-defect states and carrier dynamics in PVSK thin films, PL, time-resolved PL (TRPL), and electrical characterization were performed. The plasma-treated films displayed a 1.6-fold enhancement in steady-state PL intensity and an extended carrier lifetime from 1740.2 (Ref.) to 2751.6 ns (Target) (fig. S22), indicative of reduced defect concentration and suppressed nonradiative recombination after the plasma processing (35). Furthermore, TRPL measurements were performed on PVSK films of varying thicknesses (330, 700, and 990 nm), and the resulting data were fitted to extract the surface recombination velocities (36, 37) of 7.0 and  $14.0 \text{ cm s}^{-1}$  for Target and Ref., respectively (fig. S23).

Drive-level capacitance profiling (DLCP) (Fig. 2, A and B) on the Target device showed an order-of-magnitude reduction in defect density at the PVSK/ $\text{C}_{60}$  interface from  $6.63 \times 10^{16}$  to  $5.01 \times 10^{15} \text{ cm}^{-3}$ . Space-charge-limited current measurements showed that the trap filled limit voltage ( $V_{\text{TFL}}$ ) was reduced from 1.29 V (Ref.) to 1.06 V (Target) for the hole-only device and 1.68 V (Ref.) to 0.66 V (Target) for the electron-only device, respectively, validating the effectiveness of plasma processing in mitigating trap states (fig. S24) (38). When incorporated into full devices, the Target films exhibited a reduced dark current density ( $3.56 \times 10^{-6} \text{ A cm}^{-2}$  at  $-0.2 \text{ V}$ ) relative to the Ref. sample ( $8.42 \times 10^{-6} \text{ A cm}^{-2}$  at  $-0.2 \text{ V}$ ) (fig. S25) (39).

In Ar-based plasma, ions accelerated across the sheath typically impinged on the surface with energies on the order of  $10^1$  to  $10^2 \text{ eV}$  (40). For comparison, in thermal evaporation, the incident species



**Fig. 2. Interface defect elimination and device efficiency enhancement.** (A and B) Defect density profiles at varying depths derived from DLCP. The 0-nm interface denotes the C<sub>60</sub>/PVSK interface. (C and D) Statistical photovoltaic performance (PCE distribution) and current density–voltage (*J*–*V*) curve of the champion device fabricated with and without the plasma-assisted process, respectively.

normally carried <1 eV kinetic energy (41). Thus, species produced in the plasma surface treatment could interact with the PVSK bulk. Extended x-ray absorption fine structure (EXAFS) measurements (figs. S26 and S27 and table S3) revealed a marginal decrease in the Debye-Waller factor ( $\sigma^2$ ) from 0.021 to 0.019 after plasma treatment, indicating that the plasma process did not induce bulk structural damage.

In EXAFS analysis,  $\sigma^2$  reflects both thermal disorder and static disorder associated with lattice distortion and structural defects. Because all measurements were conducted under room-temperature conditions, the observed change in  $\sigma^2$  was primarily attributed to variations in static disorder. The slight reduction in  $\sigma^2$  suggested a more homogeneous Pb–X coordination environment with suppressed lattice distortion and reduced defect-related disorder. Although an ultrathin Pb–F-coordinated surface heterostructure could introduce local coordination heterogeneity that increases  $\sigma^2$ , the observed  $\sigma^2$  reduction in EXAFS may associate with the plasma-assisted surface “polishing” and passivation. By removing or passivating the defect-enriched surface layer, this process reduced defect-induced static disorder and decreased  $\sigma^2$  (42–44).

The electron-phonon coupling in PVSK films was further assessed through the Huang-Rhys factor (*S*) (fig. S28), which could be obtained as  $\text{FWHM} = 2.36 \sqrt{S \hbar \omega_{\text{phonon}}} \sqrt{\cot\left(\frac{\hbar \omega_{\text{phonon}}}{2k_B T}\right)}$ . In this expression, *S* denotes the electron-phonon coupling strength (Huang-Rhys factor),  $\hbar \omega_{\text{phonon}}$  the phonon frequency, FWHM the full-width of half-maximum for the PL signal peak, and  $k_B$  the Boltzmann constant (45). The Huang-Rhys factor of the Target sample was  $0.81 \pm 0.07$ , versus  $1.95 \pm 0.01$  for Ref., which indicated a reduced electron-phonon coupling strength and defect-related lattice distortions (46–48).

Previous studies demonstrated that electron-phonon interactions and the degree of crystal disorder played a crucial role in shaping the Urbach tail (49). The optical absorption of the PVSK films was further

investigated to evaluate the Urbach energies derived from fitting the UV-visible spectra (fig. S29). Plasma treatment could reduce the Urbach energy from 17.6 to 12.8 meV, demonstrating its effectiveness in decreasing electronic disorder within the absorber layer (50).

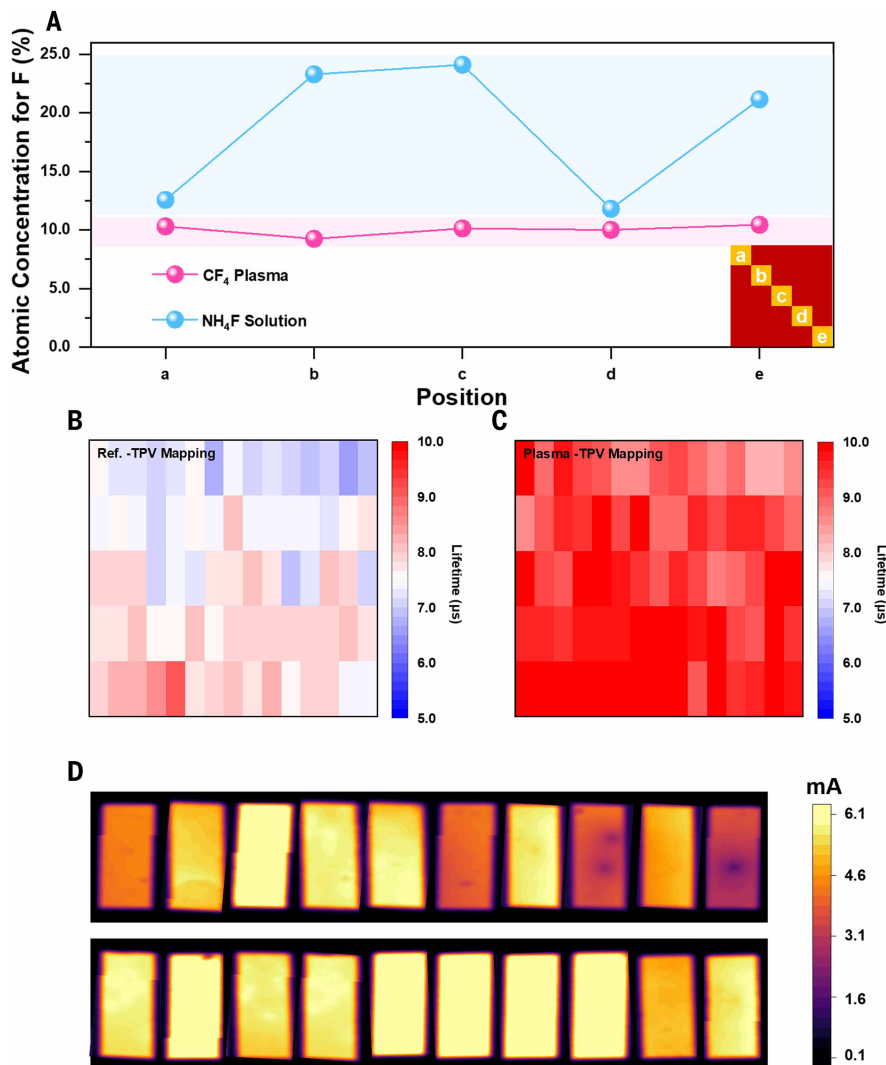
In addition to defect passivation, UV photoelectron spectroscopy (UPS) showed that the plasma treatment made the PVSK top surface more *n*-type. For the Ref., the conduction-band minimum (CBM), valence-band maximum, and Fermi level (*E<sub>F</sub>*) were located at –3.65, –5.19, and –4.70 eV, respectively. After plasma treatment, these values shifted to –3.71, –5.25, and –4.63 eV. Notably, *E<sub>F</sub>* moved upward by 0.07 eV and was closer to the CBM. Thus, the modified surface energetics improved PVSK/C<sub>60</sub> energy-level alignment (fig. S30) (51).

## Device measurements

The concurrent defect passivation and band-structure tuning collectively improved the device efficiency (supplementary text S2). The plasma-treated device (MgF<sub>2</sub>/fluorine-doped tin oxide (FTO)/self-assembled monolayer (SAM)/PVSK/C<sub>60</sub>/bathocuproine (BCP)/Ag) exhibited a champion PCE of 27.2%, an open circuit voltage (*V<sub>oc</sub>*) of 1.19 V, a short-circuit current density (*J<sub>sc</sub>*) of 26.8 mA cm<sup>-2</sup>, and a fill factor (FF) of 85.2%, compared with a PCE of 26.4%, *V<sub>oc</sub>* of 1.18 V, *J<sub>sc</sub>* of 26.7 mA cm<sup>-2</sup>, and FF of 83.5% for the Ref. device (Fig. 2, C and D, and fig. S33). The device hysteresis behavior was demonstrated in fig. S34. The *V<sub>oc</sub>* enhancement was reflected in the quasi-Fermi level splitting (QFLS), a metric that quantifies the upper limit of *V<sub>oc</sub>* governed by intrinsic nonradiative recombination losses. The Target film exhibited an increased QFLS of 1.21 eV compared with 1.19 eV for the Ref. sample (fig. S35). The external quantum efficiency results showed that the integrated current densities of the Ref. and Target samples were 26.0 and 26.2 mA cm<sup>-2</sup>, respectively, with a discrepancy of <5% compared with the measured *J<sub>sc</sub>* (fig. S36). Notably, the proposed method is also well compatible with carbon-electrode devices (fig. S37).

To evaluate the spatial uniformity and reproducibility of the plasma treatment, the element distribution and chemical state of the plasma-induced passivation layer was further examined. In this context, the F element distribution on PVSK surfaces, passivated either by spin coating an NH<sub>4</sub>F precursor solution or by plasma treatment, was compared. XPS at five diagonal positions (a through e) showed uniform F content for the plasma-treated surface (9.4 to 10.5%) but pronounced inhomogeneity for the solution-passivated sample (11.6 to 24.2%) (Fig. 3A). Atomic force microscopy and Kelvin probe force microscopy (KPFM) were performed (figs. S38 and S39). The plasma treatment induced a slight increase of surface roughness (Ref. ~14 nm and Target ~18 nm). However, KPFM mapping revealed that plasma treatment induced a ~490-mV increase (the numerical difference from UPS may be related to the different measurement environments, ambient versus vacuum) in mean contact potential difference, indicating a shift toward *n*-type properties—along with a reduction in surface potential heterogeneity—as the FWHM narrowed from 280 mV (Ref.) to 30 mV (Target). These results indicated that the process could effectively suppress surface defects while improving interfacial electrical uniformity (52, 53).

The homogenized surface energetics translated into improved carrier dynamics across the entire active area. In this aspect, TRPL mapping results revealed that the carrier lifetime of the plasma-treated samples increased from 95.9 to 146.8 ns, with the distribution variance decreasing from 6.66 to 5.59 ns (figs. S40 and S41), which indicated inhibited trap-mediated nonradiative recombination throughout the



**Fig. 3. Surface treatment uniformity and effectiveness.** (A) Spatial distribution of fluorine on passivated PVSK surfaces probed by XPS. After solution-based or plasma passivation, XPS measurements were performed at five positions (a to e) selected along the diagonal of the sample, revealing the fluorine-content distribution across the surface. (B and C) The TPV mapping of devices with or without plasma treatment, respectively. The sample size was 2 mm by 2 mm. (D) LBIC measurements of 10 randomly selected subcells from the module. The subcells were separated from a prescribed module substrate after the treatment process, followed by electrode evaporation to facilitate characterization. The upper panel shows the conventional solution-treated group, and the lower panel corresponds to the plasma-treated group.

PVSK film (54). The homogeneous electrical behavior, combined with suppressed nonradiative recombination, collectively contributed to a promoted QFLS distribution (Target: average 1.21 eV, variance 0.88 meV; Ref.: 1.17 eV, variance 0.98 meV) (fig. S42) (55). For the complete device, transient photovoltage (TPV) mapping demonstrated a general enhancement in the lifetime of the Target compared with the Ref. (Fig. 3, B and C), corresponding to the minimized detrimental recombination loss after plasma processing all through the active area (56).

To further evaluate this spatial consistency, two types of large-area (10 cm by 10 cm) devices were fabricated: One was passivated by plasma treatment and the other by conventional solution-based passivation (1 mg ml<sup>-1</sup> phenformin hydrochloride in 2,2,2-trifluoroethanol), and each device was then divided into 25 subcells to test their PCE (fig. S43). The plasma-passivated devices exhibited subcell PCEs predominantly within the 22 to 24% range. By comparison, subcells passivated

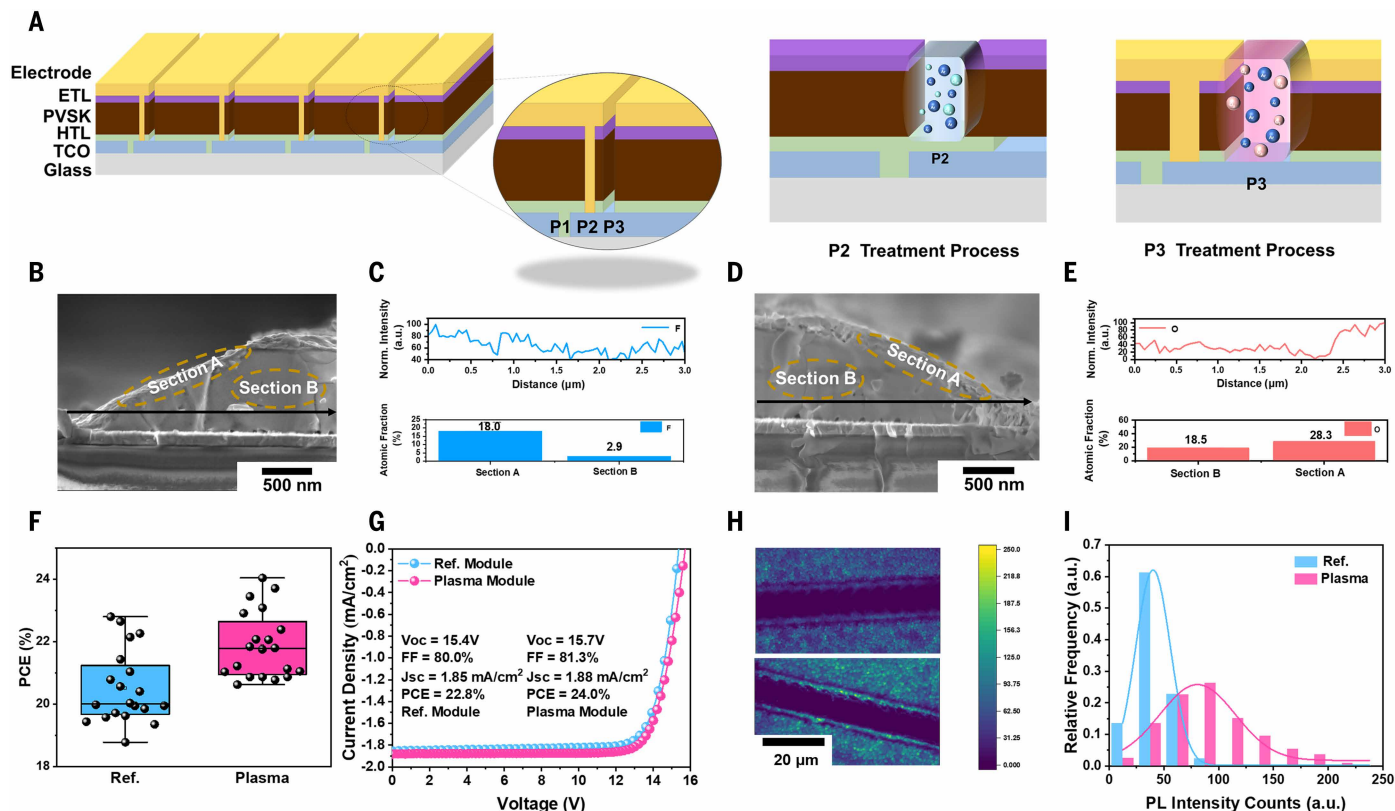
with the solution precursor displayed a broader PCE spread of 18 to 24%. We suggest that this discrepancy arose from surface inhomogeneity and reconstruction caused by solvents in the solution-based passivation method (24). Moreover, laser beam-induced current (LBIC) measurements were consistent with the subcell PCE distribution (Fig. 3D), revealing higher intensity and improved spatial uniformity in plasma-treated devices (mean: 5.5 mA; variance: 0.56 mA) compared with solution-passivated references (mean: 4.2 mA; variance: 1.4 mA).

Plasma surface treatment mitigated adverse defects within the HAZ (P2 and P3 laser-scribing zones) in modules (Fig. 4A). Consistent with the top-surface treatment, the HAZ region was optimized using Ar diluted with a functional gas. Notably, the optimal plasma chemistry was interface-specific: Considering the different carrier dynamics near the HAZ compared with the film top surface, we evaluated the potential of other functional gases, including O<sub>2</sub>, and found that O<sub>2</sub> was more effective in the P3 region, whereas CF<sub>4</sub> was preferred in the P2 region (supplementary text S3).

We used cross-sectional scanning electron microscopy (SEM) and EDX analyses to examine the distribution of characteristic elements. Specifically, in the P2 HAZ region, CF<sub>4</sub> plasma (Fig. 4, B and C) raised the surface F atomic ratio from the bulk level of 2.9% to 18.0%, whereas in the P3 region, O<sub>2</sub> plasma (Fig. 4, D and E) increased the surface O atomic ratio from the bulk level of 18.5% to 28.3%. The relatively high O content detected in the bulk region could have been caused by surface-adsorbed oxygen or surface oxidation of PVSK during annealing in air (57, 58).

Device optimization was initially performed on minimodules (aperture area, 0.75 cm<sup>2</sup>) comprising three subcells. The statistical results showed that for devices treated with the plasma process after P2 scribing, the average values of PCE,  $V_{oc}$ , FF, and  $J_{sc}$  improved from 23.1%, 3.56 V, 80.5%, and 8.04 mA cm<sup>-2</sup> to 23.4%, 3.57 V, 81.3%, and 8.05 mA cm<sup>-2</sup>, respectively (figs. S44A and S45). The PCE for the champion device increased from 23.6 to 23.8%. For devices that underwent plasma treatment after P2 and P3 scribing, the average values of PCE,  $V_{oc}$ , FF, and  $J_{sc}$  improved from 23.0%, 3.56 V, 80.6%, and 8.02 mA cm<sup>-2</sup> to 24.3%, 3.62 V, 82.4%, and 8.14 mA cm<sup>-2</sup>, respectively, with the PCE for the champion device rising from 23.1 to 24.7% (figs. S44B and S46).

The optimized process was subsequently applied to 10 cm by 10 cm modules with an aperture area of 65.05 cm<sup>2</sup>. The statistical results revealed that for modules treated with plasma in the HAZ, the average values of PCE,  $V_{oc}$ , FF, and  $J_{sc}$  improved from 20.5%, 15.0 V, 71.8%, and 1.85 mA cm<sup>-2</sup> to 21.9%, 15.2 V, 76.9%, and 1.86 mA cm<sup>-2</sup>, respectively. The efficiency for the champion device increased from 22.8 to 24.0% (Fig. 4, F and G, and fig. S49). One of the best devices was certified by the National Institute of Metrology, China, with efficiencies of 23.5% (reverse scan), 22.7% (forward scan), and 23.1% (stabilized PCE) (figs. S50 and S51). The enhanced device performance stemmed from the elimination of HAZ-specific defects. Specifically, plasma treatment led a twofold increase in fluorescence signal (40.1 to 80.7 counts) (Fig. 4, H and I). This localized remediation quenched nonradiative



**Fig. 4. Performance and implementation of the plasma treatment process toward large-area modules.** (A) Schematic illustration of plasma processing applied to treat HAZ. (B and C) Cross-sectional SEM morphology of the HAZ after CF<sub>4</sub> plasma treatment, along with the line scan (dark arrow) and selected area EDX signal distribution of the F element. (D and E) The corresponding results for O<sub>2</sub> plasma treatment. (F and G) Statistical PCE distribution and J-V curves of the champion device (65.05-cm<sup>2</sup> module) with and without plasma treatment after P2 and P3 scribing. (H and I) Confocal fluorescence microscopy images in the HAZ with (bottom) and without (top) plasma treatment, together with the corresponding intensity statistics.

recombination pathways and optimized carrier transport across the interconnect regions (59).

### Device durability

In situ heating Fourier transform infrared spectroscopy (FTIR) measurements revealed that the formamidinium (FA<sup>+</sup>) vibrational features (for a typical FAMACsPbIBr PVSK composition, where MA is methylammonium) at 1709 cm<sup>-1</sup> persisted up to ~300°C, versus a limit of ~280°C for the Ref. (figs. S52 and S53). This result underscored the role of strong hydrogen bonding in suppressing organic component degradation. In situ heating differential electrochemical mass spectrometry (DEMS) results showed that the Ref. PVSK film initiated decomposition, evidenced by triazine formation, at 147°C, whereas the Target film delayed the onset of decomposition until 158°C (fig. S54). A similar trend was also observed when the composition was shifted to the more volatile MAPbI<sub>3</sub> (fig. S55) and when the films were characterized on a larger scale (fig. S56).

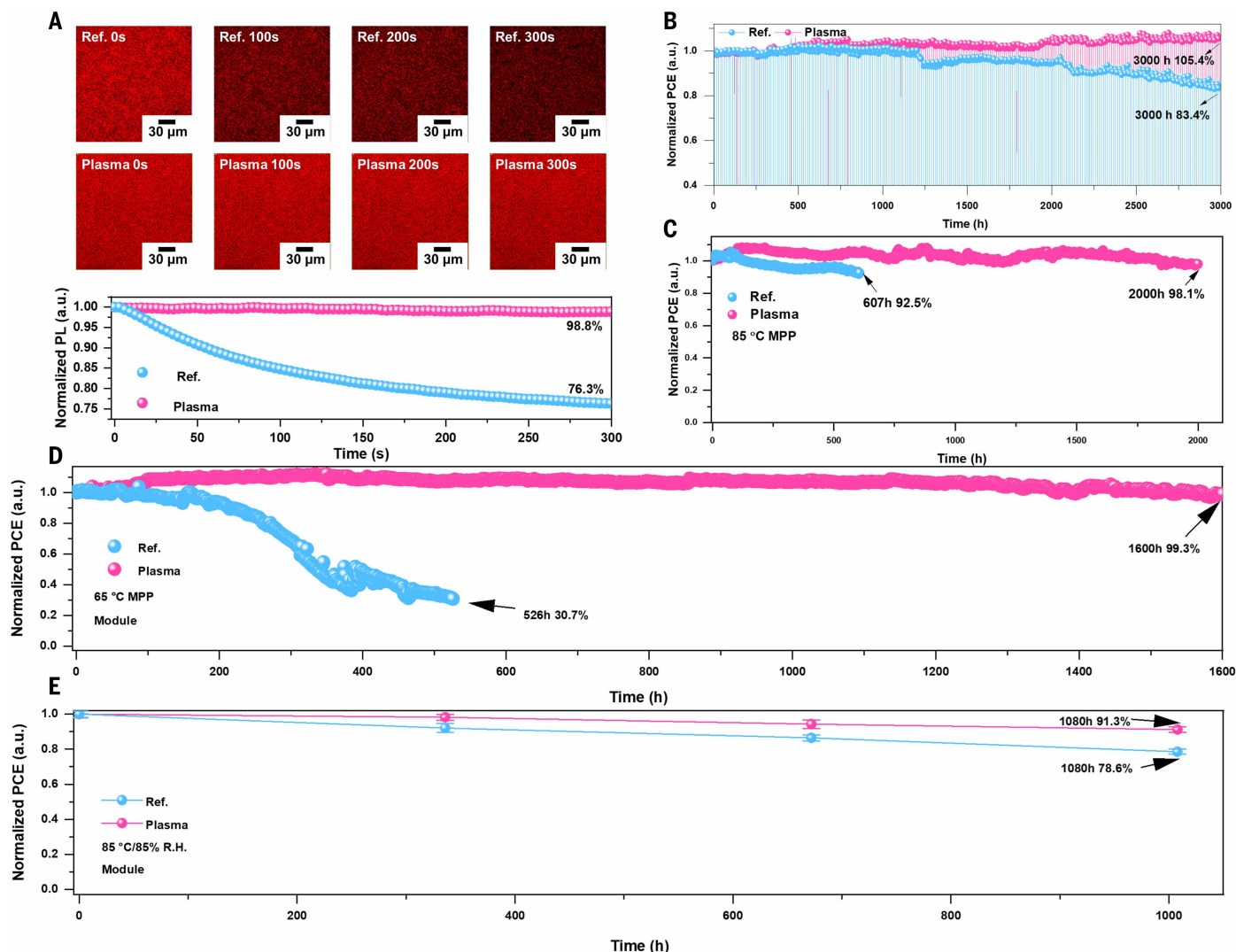
The plasma treatment also inhibited the formation of PbI<sub>2</sub> under thermal stress according to the XRD results (fig. S57). To further probe plasma-induced changes in the inorganic lattice, Raman mapping was performed, focusing on the 82.2 cm<sup>-1</sup> Pb-I stretching-band intensity (figs. S58 and S59). After aging at 85°C in air with 60 to 80% relative humidity (RH), the Raman intensity dropped 17% for the Ref. film but only 6% for the Target film (60, 61). Furthermore, the ion migration activation energy of the film increased from 0.07 to 0.21 eV (fig. S60), a result consistent with the trend of PL or TRPL variation under bias (Fig. 5A and fig. S61). These effects arose because (i) strong Pb-F ionic

bonding and F...H interactions with organic moieties stabilized the lattice and suppressed phase degradation or ion migration (62); (ii) the intrinsic hydrophobicity of F enhanced resistance to moisture and oxygen; and (iii) a continuous, uniform fluoride heterostructure formed on the top surface.

Reverse-bias breakdown tests showed that plasma-treated devices exhibited a higher average breakdown voltage, increasing from 9.5 to 14.3 V (fig. S62), which we attributed to the presence of dense, continuous F-containing heterostructure (63). We also subjected the PVSK films to continuous 365-nm UV irradiation at an intensity of ~10 mW cm<sup>-2</sup> for different exposure durations. After the designated UV aging times, the charge-transport layers and electrodes were sequentially deposited to complete the full-device fabrication. The Target devices retained 94.8% of their initial PCE after 1176 hours (117.6 kWh m<sup>-2</sup>, around 7.8 times that of the International Electrotechnical Commission (IEC) standard 61215 UV preconditioning dose of 15 kWh m<sup>-2</sup>), whereas the Ref. devices decreased to 84.6% after only 672 hours (fig. S63).

Under 85°C N<sub>2</sub> aging, the Target device showed no appreciable decay after 3192 hours, whereas the Ref. retained only 69.4% of its initial PCE after 3024 hours (fig. S64). For unencapsulated devices stored under vacuum (~100 Pa), the Target devices remained stable for 3024 hours, whereas the Ref. sample dropped to 88.1% after 1344 hours (fig. S65). Under light-dark (12 hour/12 hour) cycling, the Target devices outperformed the Ref. sample (105.4% versus 83.4% at 3000 hours; Fig. 5B).

For the maximum power point (MPP) test, Target devices retained 98.1% of their initial PCE after 2000 hours of continuous operation at 85°C in N<sub>2</sub> under 1-sun illumination, whereas Ref. devices retained 92.5%



**Fig. 5. Stability evaluation of the plasma-treated samples.** (A) In situ bias-modulated PL evolution with an ITO/PVSK/Au structure under 1.2 V and corresponding PL mapping images of different stages. (B) Tracking performance over a 24-hour diurnal cycle (12 hours light/12 hours dark) for the Ref. and Target devices. (C) Device MPP test results at 85°C under 1-sun illumination in a  $\text{N}_2$  environment. (D) Solar module MPP test results at 65°C under 1-sun illumination in air with 60 to 80% relative humidity. (E) Damp-heat tests for encapsulated modules at 85°C and 85% RH.

after 607 hours (Fig. 5C). The 10 cm by 10 cm Target modules with encapsulation aged at 65°C in air with 60 to 80% RH under 1-sun illumination retained 99.3% after 1600 hours, whereas Ref. modules retained 30.7% after 526 hours (Fig. 5D). When the temperature was raised to 85°C for the MPP test, the Target module retained 87.2% of its initial PCE after 1028 hours, whereas the Ref. retained only 64.9% (fig. S66). Under IEC 61215:2016 damp-heat testing (85°C and 85% RH), Target modules maintained 91.3% after 1080 hours versus 78.6% for Ref. modules (Fig. 5E).

## Conclusions

Plasma-mediated surface chemical mechanisms enabled precise control of interfacial heterostructure and charge-carrier dynamics in the soft PVSK lattice, thereby overcoming the long-standing trade-offs between device performance and scalability in high-efficiency PSCs. The combination of Ar-driven physical etching and in situ  $\text{CF}_4$ -assisted fluorination reconstructed the defect-enriched surface into a stabilized Pb-F-coordinated heterostructure that suppressed lattice distortion and mitigated defect-assisted degradation pathways. This plasma

process was applicable not only to pristine film surfaces but also to laser-induced HAZs in modules. As a result, small-area cells achieved a PCE of 27.2%, and 10 cm by 10 cm modules (aperture area of 65.05  $\text{cm}^2$ ) reached a PCE of 24.0%. More notably, under MPP operation at 85°C with continuous illumination, plasma-processed devices maintained 98.1% of their initial PCE after 2000 hours. At the module level, encapsulated 10 cm by 10 cm devices preserved 99.3% of their initial PCE after 1600 hours at 65°C with 1-sun continuous illumination. Additionally, the plasma-engineered interfaces imparted reliable resistance toward light-dark cycling (3000 hours), thermal stress (3192 hours), UV irradiation (117.6  $\text{kWh m}^{-2}$ ), reverse-bias operation ( $-14.3$  V), and low-pressure environments ( $\sim 100$  Pa).

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## SUPPLEMENTARY MATERIALS

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Materials and Methods; Supplementary Text S1 to S3; Figs. S1 to S66; Tables S1 to S3; References (64–93)

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## Plasma surface engineering for efficient and stable perovskite solar cells and modules

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### Editor's summary

A dry plasma etching process forms perovskite-lead fluoride heterostructures and passivates defects across large areas while maintaining bulk film integrity. Fan *et al.* showed that this process decreases surface defects by an order of magnitude and could also be applied to laser-scribed interconnection regions that dominate module-level performance loss. Small-area devices achieved a champion power conversion efficiency (PCE) of 27.2%, and 100-square-centimeter modules achieved a champion PCE of 24.0%. When incorporated into the stabilized device structure, small-area devices retained 98.1% of their initial PCE after 2000 hours of maximum power point tracking (MPPT) at 85°C, and 100-square-centimeter modules retained 99.3% of their initial PCE after 1600 hours of MPPT at 65°C. —Phil Szuromi

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